

Approximation Algorithms

Exam-Ready Compact Notes

1 Core Idea

Approximation Algorithm

For a hard optimization problem, we accept a **near-optimal** solution that can be found in polynomial time.

Exact algorithm	Finds the optimum, but may be too slow.
Approximation algorithm	Polynomial time + guaranteed quality.
Good ratio	Closer to 1 is better.

2 Approximation Ratio

Let C be the algorithm's cost and C^* the optimal cost.

Definition

$$\max\left(\frac{C}{C^*}, \frac{C^*}{C}\right) \leq \rho(n)$$

Then the algorithm is a $\rho(n)$ -approximation.

Minimization	$C \leq \rho C^*$
Maximization	$C \geq \frac{C^*}{\rho}$
2-approximation	Solution is within a factor of 2 of optimum.

PTAS

A PTAS takes an accuracy parameter $\varepsilon > 0$ and gives ratio $1 + \varepsilon$ in polynomial time for every fixed ε .

3 Vertex Cover

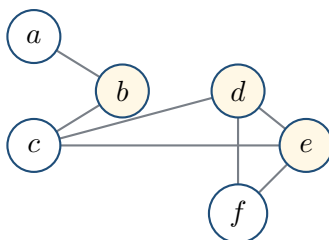
Definition

A set $C \subseteq V$ is a vertex cover if every edge (u, v) has $u \in C$ or $v \in C$. Goal: minimize $|C|$.

3.1 Greedy 2-Approximation

Algorithm

1. $C \leftarrow \emptyset$.
2. Pick any uncovered edge (u, v) .
3. Add both u and v to C .
4. Delete all edges incident on u or v .
5. Repeat until no edge remains.



Why It Is a 2-Approximation

Let A be the edges selected by the algorithm.

1. Selected edges share no endpoint, so A is a matching.
2. Any vertex cover must take at least one endpoint from each edge in A .

$$|C^*| \geq |A|$$

3. The algorithm takes both endpoints of every selected edge.

$$|C| = 2|A|$$

Therefore,

$$|C| = 2|A| \leq 2|C^*|$$

So the algorithm is a 2-approximation.

Time

With adjacency lists and marking deleted edges:

$$O(|V| + |E|)$$

4 Metric Traveling Salesperson Problem

Metric TSP

Find a minimum-cost Hamiltonian cycle in a complete graph where edge costs satisfy:

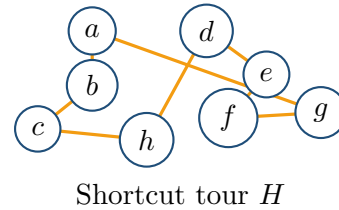
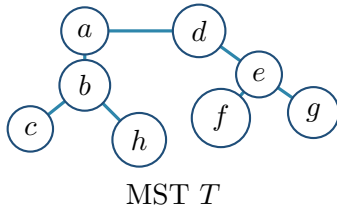
$$c(u, w) \leq c(u, v) + c(v, w)$$

This is the triangle inequality.

4.1 2-Approximation Algorithm

Algorithm

1. Compute a minimum spanning tree T .
2. Walk around T ; every tree edge is used twice.
3. Record vertices in preorder.
4. Skip repeated vertices (shortcut them).
5. Return to the start.



Why It Is a 2-Approximation

Let H^* be an optimal TSP tour.

1. Remove one edge from H^* ; what remains is a spanning tree. Since T is minimum:

$$\text{cost}(T) \leq \text{cost}(H^*)$$

2. Walking around T uses every tree edge twice:

$$\text{cost}(W) = 2 \text{cost}(T) \leq 2 \text{cost}(H^*)$$

3. Shortcut repeated vertices. Triangle inequality guarantees no increase in cost:

$$\text{cost}(H) \leq \text{cost}(W)$$

Hence,

$$\boxed{\text{cost}(H) \leq 2 \text{cost}(H^*)}$$

So the algorithm is a 2-approximation.

Time

Using Prim's algorithm on a complete graph:

$$\boxed{O(|V|^2)}$$

5 Why Triangle Inequality Is Necessary

General TSP

If $\mathbf{P} \neq \mathbf{NP}$, general TSP has no polynomial-time constant-factor approximation.

Compact Proof

Assume a ρ -approximation exists for general TSP. Given graph G with n vertices, build a complete graph with

$$c(u, v) = \begin{cases} 1, & (u, v) \in E, \\ \lceil \rho n \rceil + 1, & (u, v) \notin E. \end{cases}$$

- If G has a Hamiltonian cycle, optimum TSP cost is n , so the approximation returns cost at most ρn .
- If G has no Hamiltonian cycle, every TSP tour uses at least one expensive non-edge, so its cost is greater than ρn .

Therefore we could decide Hamiltonian Cycle in polynomial time by checking whether the returned cost is at most ρn . That would imply $\mathbf{P} = \mathbf{NP}$.

6 Final Memory Sheet

Topic	One-line memory
Approximation ratio	Compare returned cost with optimum; closer to 1 is better.
PTAS	Ratio $1 + \varepsilon$ for fixed ε .
Vertex Cover	Pick an uncovered edge, take both endpoints.
VC proof	Matching gives $ C^* \geq A $ and algorithm gives $ C = 2 A $.
Metric TSP	MST $\rightarrow$ double walk $\rightarrow$ shortcut.
TSP proof	MST $\leq OPT$, doubled MST $\leq 2OPT$, shortcut never increases cost.
General TSP	No constant-factor approximation unless $P = NP$.